

Tanvir Ahmed

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RESEARCH INTERESTS

Cyber Security and Intrusion Detection • On-Device LLM and Vision-Language Models • Trustworthy and Calibrated Machine Learning • Quantum Machine Learning • Medical Image Analysis

RESEARCH SUMMARY

I am a computer science graduate and independent researcher. My work covers network security, medical image analysis and quantum machine learning, and in all three I focus on whether the results hold up. Four of my papers were accepted in 2026. One of them, at ICTAI, shows that feature selection can leave overall accuracy almost unchanged while making a rare attack class far harder to detect, and gives a method that finds the harmed classes and repairs the subset. A related line treats intrusion-detection feature selection as a combinatorial optimization problem, solved with QAOA and with simulated annealing, then benchmarked against classical selectors on industrial IoT traffic (X-IIoTID, NF-ToN-IoT, NF-UNSW-NB15). I also co-authored an accepted agent that pairs a vision-language model with an LLM and runs both locally, so web pages are screened on consumer hardware without anything leaving the device, and two accepted papers on trustworthy tuberculosis microscopy. Most of my effort goes into evaluation that holds up: patient-disjoint splits, leakage audits and per-class results rather than a single headline score.

EDUCATION

Independent University, Bangladesh

January 2015 - December 2019

Department of Computer Science and Engineering

Bachelor of Science in Computer Science

CGPA: 2.92 out of 4.00 (Final-year GPA: 3.07)

PUBLICATIONS

Conference papers

Accepted

- **Tanvir Ahmed**, Jannatul Ferdous Shorna, Md Mahbubur Akash, Md Khorshed Alam, and Md Tarek Hasan. “DARE-FS: A Selector-Agnostic Operator for Detecting and Repairing Class-Specific Feature-Selection Harm.” *2026 IEEE 38th International Conference on Tools with Artificial Intelligence (ICTAI)*, 2026. Code: [GitHub].
- Md Faisal Sheikh, **Tanvir Ahmed**, Enam Ahmed Taufik, Abhijit Kumar Ghosh, Mohammad Zavid Parvez, Prabal Barua, and Subrata Chakraborty. “Beyond Confidence: A Leakage-Audited Selective-Prediction Study of Acid-Fast Bacilli Detection in Ziehl-Neelsen Microscopy.” *2026 International Conference on Digital Image Computing: Techniques and Applications (DICTA)*, 2026. Code: [GitHub].
- Swagotam Malakar, **Tanvir Ahmed**, Abhijit Kumar Ghosh, Mohammad Zavid Parvez, Prabal Barua, and Subrata Chakraborty. “Towards Trustworthy Tuberculosis Detection: Subgroup Calibration of Object Detectors in Sputum-Smear Microscopy.” *2026 International Conference on Digital Image Computing: Techniques and Applications (DICTA)*, 2026. Code: [GitHub].
- Dhiman Kumar Sarker, Fyaz Nafin Rahman, Md Rezaul Karim Khan, Sushmoy Paul Rohit, **Tanvir Ahmed**, and Md Mizanur Rahman. “Real-Time Browser Threat Detection Security Agent via Local VLM-LLM.” *2026 5th IEEE International Conference on Robotics, Automation, Artificial Intelligence and Internet-of-Things (RAAICON)*, 2026. Code: [GitHub].

Journal paper

Under Review

- Abhijit Kumar Ghosh, Ar Rafiul Islam Rafim, Fahmid Al Farid, **Tanvir Ahmed**, Enam Ahmed Taufik, Jia Uddin, and Hezerul Abdul Karim. “Gated Dual-Scale Feature Fusion for Colorectal Cancer Grading in Centralized and Federated Pathology.” *Informatics in Medicine Unlocked* (Elsevier), 2026. Code: [GitHub].

PROJECTS

QCIDS: Quantum-Classical Feature Selection for Network Intrusion Detection [GitHub]

- Cast intrusion-detection feature selection as a QUBO solved by QAOA ($p = 1, 2, 3$) on simulators, by simulated annealing, and by classical baselines under matched leakage-free cross-validation; tightening the budget from $k = 20$ to $k = 15$ on NF-ToN-IoT cost greedy mutual information 4.3 macro-F1 points while QAOA $p = 2$ held at 0.763.

CovNCG: Training-Free Dense Quantum Feature Maps [GitHub]

- Designed a training-free quantum kernel from non-commuting generator blocks (three features per two qubits) with a closed-form kernel-variance result scaling as c^4 , which turns avoidance of exponential concentration into a tuning rule; evaluated with a projected kernel and SVM on five medical datasets, with validation on quantum hardware.

Q-PlasmaNet: Hybrid Classical-Quantum Classifier for Plasma Cell Recognition [GitHub]

- Paired a frozen CNN backbone with a 4-8 qubit data re-uploading variational quantum head (PennyLane) at matched parameter budget, reaching F1 0.927 with 178 trainable parameters - statistically tied with a parameter-matched MLP (0.932) and below a 2.2M-parameter MobileNetV2 (0.949), establishing parity rather than quantum advantage.

PCMMD: Federated Learning and Knowledge Distillation for Plasma Cell Detection [GitHub]

- Benchmarked centralized, few-shot, knowledge-distillation (EfficientNet-B0 to MobileNetV3) and federated (FedAvg/FedProx, one client per patient) training for plasma-cell classification of bone-marrow crops, under patient-disjoint folds and image-hash de-duplication.

Direct-WAM: Self-Supervised Denoising for Two-Photon Calcium Imaging [GitHub]

- Proved a bias-floor theory of self-supervised objectives (per-objective error floor, ensemble bias-crossing, distillation transfer bound) and produced a corrected AI4Life CIDC25 benchmark in which a faithfully retrained DeepCAD-RT reaches +23.65 dB stSNR, while the F_0 -free Direct-WAM 3D U-Net distilled from a 13-way ensemble reaches +18.15 dB.

RESEARCH EXPERIENCE

Independent Researcher

Boca Raton, Florida, USA

Independent and collaborative research

January 2025 - Present

Research collaborators: Prof. Jia Uddin (Woosong University, South Korea); Dr. Mohammad Zavid Parvez (RMIT University, Australia); Assoc. Prof. Subrata Chakraborty (University of New England, Australia); Prof. Prabal Datta Barua (University of Southern Queensland, Australia)

- Work with international collaborators across machine learning for security, quantum optimization and medical imaging; five papers written in 2026, four accepted and one journal manuscript under review.
- Run the full experimental pipeline on each project: auditing the data for leakage, building and training the models, benchmarking against classical and quantum baselines, and reporting results over multiple seeds with confidence intervals.

PROFESSIONAL EXPERIENCE

Career Break - Relocation to the United States

Boca Raton, Florida, USA

F-2 dependent status

December 2022 - December 2024

- Relocated from Bangladesh to the United States as an F-2 dependent, a status that does not authorize employment; used the period for self-directed study in machine learning, which led into the independent research work above.

Upwork & Fiverr

Remote - Dhaka, Bangladesh

Freelance Web and Software Developer

December 2014 - November 2022

- Built and maintained websites and web applications for international clients, writing custom front-end and back-end code to each client's specification.
- Handled the ongoing side of the work: bug fixes, feature requests, content and data updates, and direct client support across an eight-year freelance practice.

IDOL Focus - A Professional Training Institute

Dhaka, Bangladesh

Web Developer Intern

September 2019 - December 2019

Supervisors: Md. Asif Bin Khaled, Lecturer, Department of Computer Science and Engineering, IUB (academic); Md. Al Mamun, IDOL Focus (industry)

- Built the Admin and Agent modules of an online job portal (*idolfocusjobs.com*) serving four user roles: role-based authentication, jobseeker and employer account management, CV database administration, job-posting approval, and payment handling.
- Implemented in PHP, JavaScript, Ajax, jQuery, and Bootstrap over MySQL; documented in an undergraduate internship report submitted toward the BSc degree.

TECHNICAL SKILLS

Security:	Network intrusion detection, IIoT traffic analysis, on-device threat detection agents
Trustworthy ML:	Calibration (ECE, Brier), selective prediction, subgroup and worst-group evaluation, leakage auditing
LLM & VLM:	Local LLM and vision-language model inference, 4-bit quantized deployment, multimodal threat detection
Programming:	Python, SQL, PHP, JavaScript
ML Frameworks:	PyTorch, Scikit-learn, XGBoost, LightGBM, TensorFlow, Ultralytics (YOLO), OpenCV, SHAP, NumPy, Pandas
Quantum ML:	PennyLane, Qiskit, D-Wave dimod, QAOA/QUBO formulation, variational and annealing solvers
Methods:	Object detection, federated learning, transfer learning, knowledge distillation, self-supervised learning
Web:	PHP, MySQL, JavaScript, jQuery, Bootstrap
Tools:	L ^A T _E X, Git, Jupyter Notebook, Google Colab
Languages:	Bangla (native), English (fluent)